UNIVERSITY OF PADOVA
DEPARTMENT OF INFORMATION ENGINEERING

Biomedical Wearable Technologies
for Healthcare and Wellbeing

Exam and Project Discussion
Q&A

A.Y. 2025-2026

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Outline

- **The theory exam**
- Project delivery instructions
- Project presentation guidelines

Exam structure

Written test of 90-minute duration:

- 4 open questions

Notes:

- The exam is in English, answers to be provided in English (English errors not penalized). Dictionary is not allowed.
- At the exam, the students cannot use any course material (slides, books, articles, etc.), nor any electronic device (e.g., smartphone, notebook, etc.). Only the pen and a calculator is allowed.
- The theory exam will cover all the theory program except the material marked as **BONUS**

Mark of the theory exam

- Open questions:
  - 0 to 5 points each
- Total score: up to 20 points

Final mark

- The written exam is passed if the student gets at least 12/20 points (which corresponds to 18/30).

- To get the final mark, students must pass both the written exam and the project discussion.

- The mark of the project is up to 12 (based on Q&A project discussion, different students in the same group can get different marks for the project).

- The vote of the theory exam accounts for 2/3 of the final mark.

$$final\ mark = \frac{2}{3} \cdot mark\ of\ the\ theory\ exam + mark\ of\ the\ project + symposium\ bonus$$

Examples of open questions

Q1. Explain the RSA algorithm

Q2. Explain the role of the embedded processing unit

Q3. Provide a description of the following HTTP methods: GET, HEAD, POST, PUT, DELETE.

Examples of open questions

Q4. Given the results of 5 SUS questionnaires collected by 5 participants evaluating an app A:

SUS_A = 65, 80, 93, 78, 54

- Calculate the confidence interval at 90% : **74 ± 14.25**
- Evaluate if an app B (evaluated by the same 5 participants) with SUS:

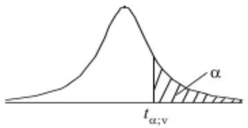
SUS_B = 90, 90, 87, 95, 91

can be said to be significantly more usable with a level of confidence >= 0.05:

t = 2.30 → NO (if it was >= 0.1, YES)

Table of the Student's *t*-distribution

The table gives the values of $t_{\alpha;\nu}$ where $\Pr(T_\nu > t_{\alpha;\nu}) = \alpha$, with ν degrees of freedom



ν \ α	0.1	0.05	0.025	0.01	0.005	0.001	0.0005
1	3.078	6.314	12.076	31.821	63.657	318.310	636.620
2	1.886	2.920	4.303	6.965	9.925	22.326	31.598
3	1.638	2.353	3.182	4.541	5.841	10.213	12.924
4	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	1.319	1.714	2.069	2.500	2.807	3.485	3.767
24	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	1.316	1.708	2.060	2.485	2.787	3.450	3.725
26	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27	1.314	1.703	2.052	2.473	2.771	3.421	3.690
28	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	1.311	1.699	2.045	2.462	2.756	3.396	3.659
30	1.310	1.697	2.042	2.457	2.750	3.385	3.646
40	1.303	1.684	2.021	2.423	2.704	3.307	3.551
60	1.296	1.671	2.000	2.390	2.660	3.232	3.460
120	1.289	1.658	1.980	2.358	2.617	3.160	3.373
∞	1.282	1.645	1.960	2.326	2.576	3.090	3.291

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General info

- In the following, we reported the instructions on how to deliver your project. In a nutshell, you must prepare and submit a **.pdf file** containing
  - the number of your group and the name of each group participant
  - the link to a public GitHub repository containing a ”valid” version of the code

- Since the project must be delivered through GitHub, code delivered via mail will be ignored

- “Valid” code must be in the **master** (or **main**) **branch** and must have been committed no later than the **12/07/2026**, and no later than the **day before the project discussion date**.

Example scenario

- ”I belong to Group 50 composed by me, Martina Vettoretti, and Luca Cossu”
- “My code is located in my public repository named bwthw”

Step 1: Fill the group info

- Create a .docx file

- Fill the group info at the very top of the file, e.g.:

Group 50

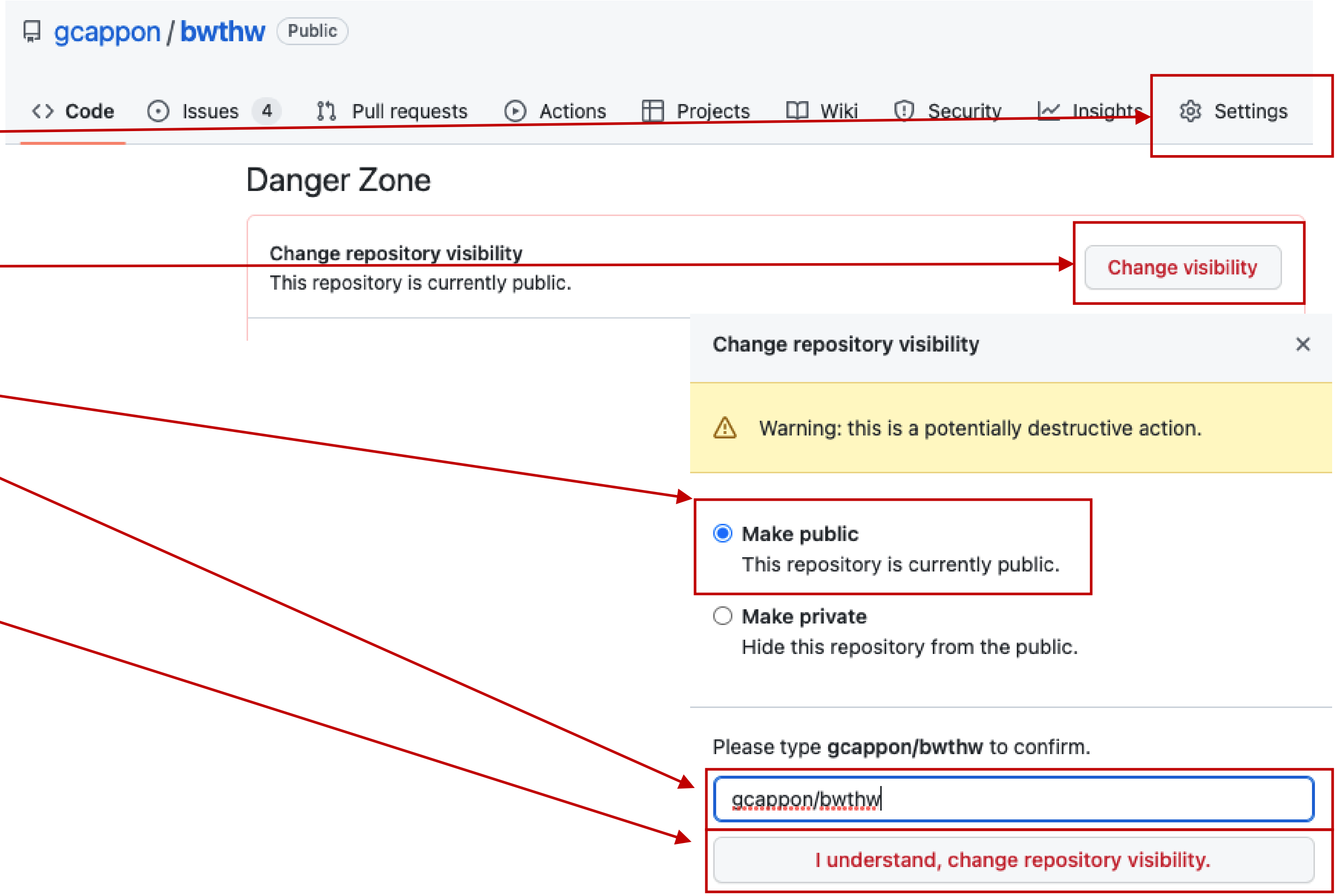
Group members:

ID	Name	Mail
1234567	Giacomo Cappon	giacomo.cappon@unipd.it
1234568	Martina Vettoretti	martina.vettoretti@unipd.it
1234569	Luca Cossu	luca.cossu@phd.unipd.it

Step 2: Make your repository public

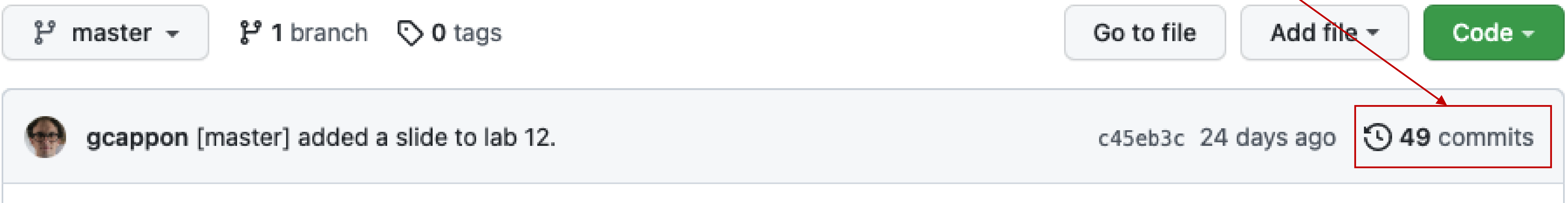
- First, make sure your repository is public. This is true if, in GitHub, a “public” label is present near your repo name:

- If, instead, the label says “private”, follow these steps to make it public:
  - Go to “Settings”
  - Scroll down till the “Danger Zone”
  - Click on “Change visibility”
  - Select “Make public”
  - Type the name of the repo to confirm
  - Click on the button

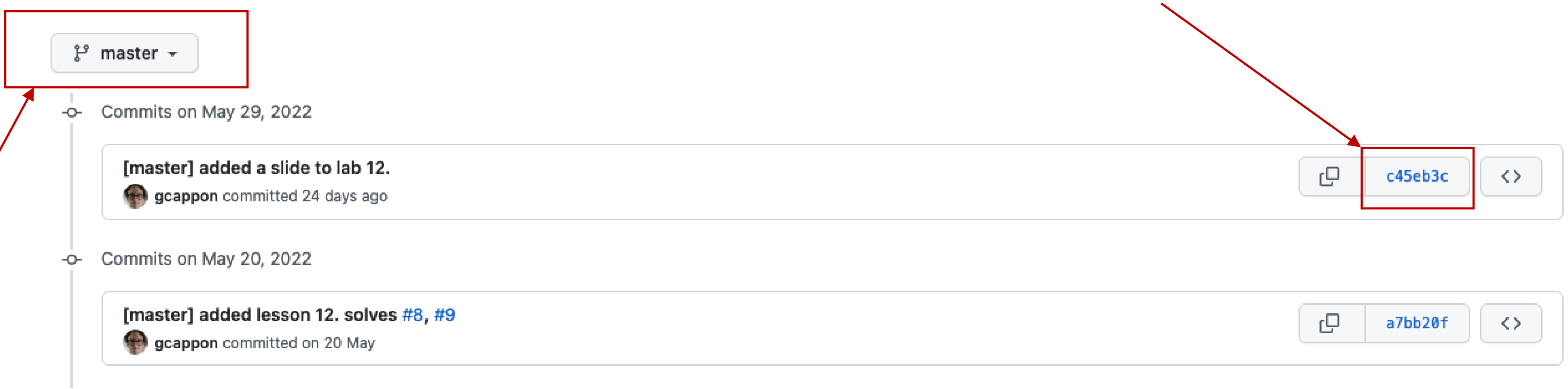


Step 3: Locate the link to a valid code version

- From the home page of your GitHub repo, click on the "history" link



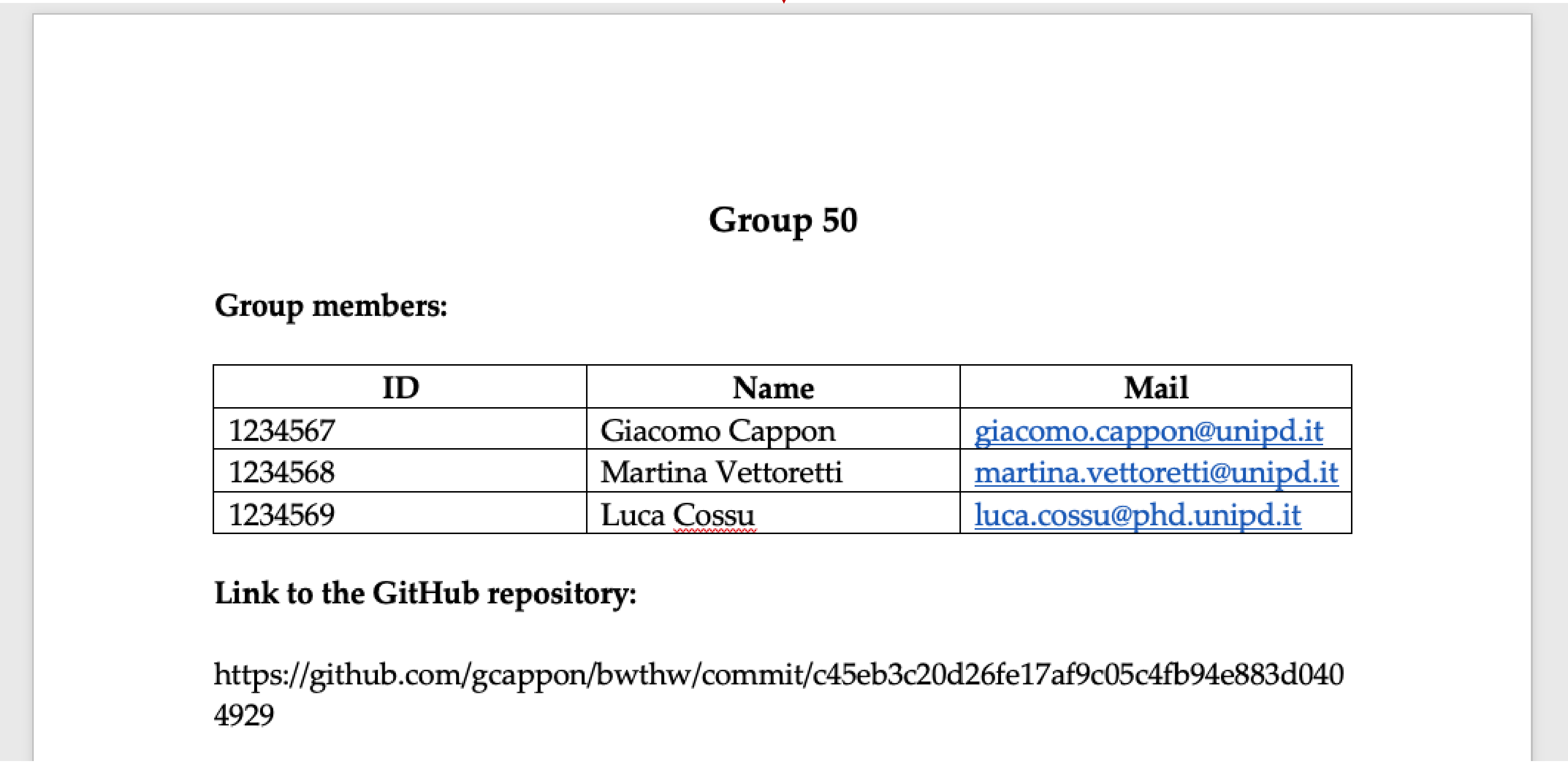
- Click on the link of the version of the code that you want to submit



- Note: be sure that your are in the master (or main) branch!

Step 4: Put the link in the .docx file

- Copy and paste the link in the url bar of your browser in the .docx file below the group info, e.g.:



Group 50

Group members:

ID	Name	Mail
1234567	Giacomo Cappon	giacomo.cappon@unipd.it
1234568	Martina Vettoretti	martina.vettoretti@unipd.it
1234569	Luca Cossu	luca.cossu@phd.unipd.it

Link to the GitHub repository:

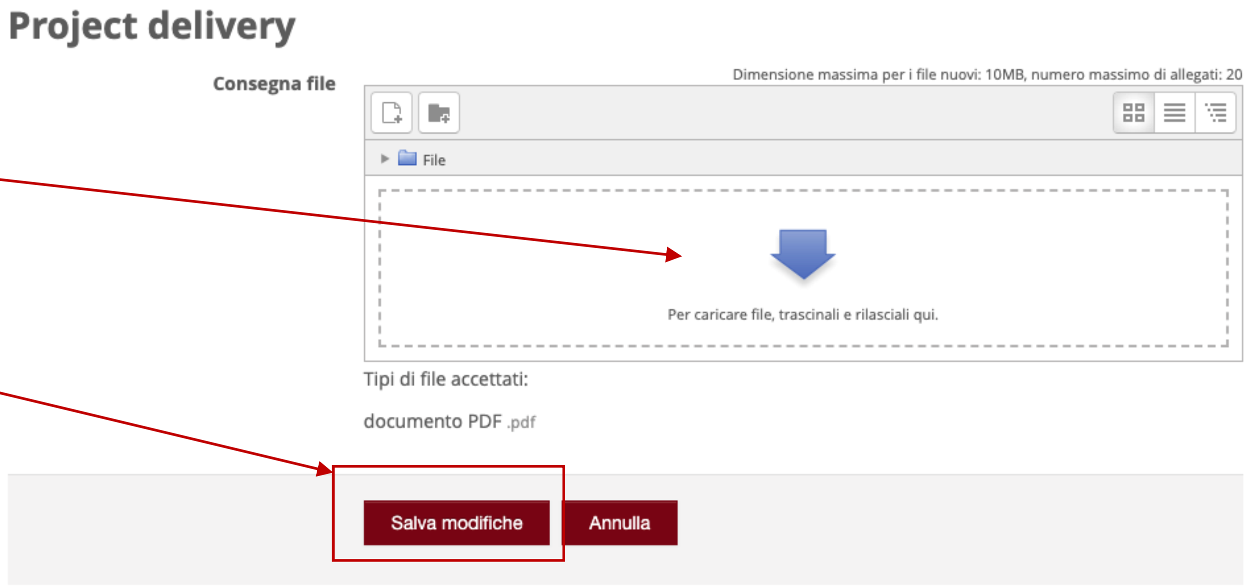
https://github.com/gcappon/bwthw/commit/c45eb3c20d26fe17af9c05c4fb94e883d0404929

Step 5: Submit the .pdf to moodle

- Export your .docx file to .pdf
- Then, go to moodle. In the “Project discussion – Project delivery” section locate “Project delivery”:

- Click on the delivery button:

- Upload the .pdf file
- Save and that’s it!



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Rationale

- In the following we reported a brief guide that contains some **hints** on what to cover during your project discussion. As such, consider this document as a guide rather than something to “fill” with words and images.
- You are free to use the template your prefer
- The skeleton proposed here is just a simple guideline. Feel free to change it according to your needs.

Discussion structure – General info

- You have **25 minutes max** to present your project followed by ~20 min of Q&A. Discussions of more than 25 minutes will be penalized.
- The presentation must include a PowerPoint presentation followed by a live demonstration of your app working. Both the presentation and the demo must fit the 25 minutes.
- The discussion must be in English.
- The Q&A session will be in English.

Discussion structure – (Possible) skeleton

- A possible skeleton for your discussion can be:
  - **Background/Target**
  - **Problem to be solved/Feature to be provided to the public –**
  - **Your solution**
  - **Core app functionalities**
  - **Original part**
  - **Project management flavours**
  - **Special implementation decisions**
  - **Closing remarks and future developments**

  - **Live demo** - *Show (via emulator or physical device) your app working.*

Discussion suggestions

- Try to motivate every choice.
- Consider in advance how much time you will allocate to the PowerPoint presentation and how much time for the live demo.
- Prepare the live demo. You have limited time so you must know what to ”tap”, to show, to do ,….
- Be sure that your app is working when you run the demo.